

SkillPulse Project Deployment Summary

Full Troubleshooting, Docker Configuration & GitHub Actions CI/CD Report

1. Project Overview & Architecture

SkillPulse ek full-stack application hai jise Docker Compose ka use karke microservices architecture par deploy kiya gaya hai. Is pure configuration aur debugging process me teen main components ko orchestrate aur automate kiya gaya:

- **Frontend Service:** Nginx server par hosting jo port 80 par client requests receive karta hai.
- **Backend Service:** Go (Golang) language me likhi gayi REST API jo port 8080 par business logic execution karti hai.
- **Database Service:** MySQL 8.0 server jo persistent storage aur data consistency handle karta hai.

2. Issues Encountered & Resolved

Deployment aur automation ke dauran jo main technical blocks aaye, unhe step-by-step resolve kiya gaya:

A. GitHub Authentication Block (Password Deprecation)

- **Issue:** Self-hosted runner par git pull origin main chalane par bar-bar Invalid username or token. Password authentication is not supported ki error aa rahi thi.
- **Resolution:** GitHub ke standard password deprecation policy ke tahat terminal auth ko bypass karne ke liye do methods select kiye gaye. Ya toh repository remote URL me **Personal Access Token (PAT Classic)** ko explicit embed kiya gaya, ya fir workflow me official actions/checkout@v4 integration ka use karne ka decision liya gaya jo token-less secure cloning handle karta hai.

B. Network Port Conflicts (Bind 0.0.0.0:8080 Failed)

- **Issue:** Docker containers up karte waqt port is already allocated ka error aa raha tha jisse Backend container "Starting" status par hung/freeze ho jata tha.
- **Resolution:** Initial state me Nginx Frontend aur Go Backend dono hi port 8080 allocate karne ki koshish kar rahe the. System ke processes clear karne ke liye docker compose down aur zombie networks ko clear kiya gaya. Final architecture me Frontend ko port 80:80 aur Backend ko port 8080:8080 par completely isolate kiya gaya.

C. Docker Compose YAML Syntax Errors

- **Issue:** Environment variables me key-value assignment ke liye = operator aur incorrect lists/objects indentation formatting errors thi.

- **Resolution:** Pure Docker Compose file ko proper structure aur specification version 3.8 ke standard par standardise kiya gaya.

3. Final Valid Docker Compose Configuration

Niche di gayi clean configuration file aapke machine par stable running state me successfully verify ki gayi hai:

```
version: '3.8'

services:
  database:
    image: mysql:8.0
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: "2002"
      MYSQL_DATABASE: skillpulse
      MYSQL_USER: rootuser
      MYSQL_PASSWORD: "2002"
    volumes:
      - mysql_data:/var/lib/mysql
    healthcheck:
      test: ["CMD-SHELL", "mysqladmin ping -h localhost -u root -p$$2002"]
      interval: 10s
      timeout: 5s
      retries: 5
      start_period: 10s

  backend:
    image: go-backend:latest
    build:
      context: ./backend
      dockerfile: Dockerfile
    restart: always
    ports:
      - "8080:8080"
    environment:
      - DB_HOST=database
      - DB_USER=rootuser
      - DB_PASSWORD=2002
      - DB_NAME=skillpulse
    depends_on:
      database:
        condition: service_healthy

  frontend:
```

```

image: nginx-frontend:latest
build:
  context: ./frontend
  dockerfile: Dockerfile
restart: always
ports:
  - "80:80"
depends_on:
  backend:
    condition: service_started

volumes:
  mysql_data:

```

4. Production Deployment & Verification Commands

Containers ke proper synchronization aur port blocking hatane ke liye sequential troubleshooting steps follow kiye gaye:

```

# 1. Stop and clean up any stuck container/zombie networks
docker compose down

# 2. Force stop any process binding to host networks if needed
docker stop $(docker ps -aq) 2>/dev/null || true

# 3. Fresh build images without using obsolete cache and start in detached mode
docker compose up -d --build

# 4. View active status and verify container health checks
docker ps

```

5. Service Mapping & Verification Status

Is setting ke baad aapka application system resources aur public networks par is tarah mapped hai:

Service Name	Internal Container Port	Exposed Host Port	Local Verification URL
Nginx Frontend	80/tcp	80	http://localhost
Go Backend API	8080/tcp	8080	http://localhost:8080/health
	3306/tcp	Isolated	Managed via Backend Network

Service Name	Internal Container Port	Exposed Host Port	Local Verification URL
MySQL Database			

Current Project State: Local runner par port isolation completely successful ho chuka hai, saare docker components stable chal rahe hain, aur infinite loading ka bug system network mapping change karne se fix ho gaya hai.